

TRACK

cloud3,521

edge891

mobile412

tinyml1,104

DIFFICULTY

L1

L2

L3

L4

L5

L6+

COMPETENCY

All areas

ZONE

diagnosis

☒ Napkin math only

☐ Visual questions only

☒ Chained questions only

847 in pool

· chains

L4SYSTEMSdiagnosiscloud

Report

PART 4 OF 6roofline-analysis · chain

Decode MFU is 8% at batch 16  
— where is the roofline?

YOUR TASK

Name the bound and show arithmetic intensity is below the ridge point.

Your production LLM serving reports ~8% MFU during decode at B≈16.  
H100: 494 TFLOPS (dense TF32) · 3.35 TB/s HBM. Team suspects underclocking.  
You think memory bandwidth — prove it with napkin math before peeking.

YOUR NAPKIN MATH

# weights ~140 GiB @ FP16 → one full pass per step reads ~140 GiB  
# AI = B \* FLOP/byte ~ 16 FLOP/B | H100 ridge ~ 148 FLOP/B  
# → memory-bound; ceiling set by 3.35 TB/s, not 494 TFLOPS...

Type final line as ⇒54e12 to check against model (%⌨)

Reveal Answer

Stuck? Ask Interviewer in the tools panel →12 in pool · Next N

Filters (left) slice the corpus; centre = commit-then-reveal loop; right = tools.

TOOLS

Skip

HARDWARE REFopen

H100 · HBM3e 3.35 TB/s  
Peak TF32: 494 T → ridge ~148 F/B  

table scrolls

Napkin calculator

ASK INTERVIEWER

Journal mode. Your clarifying questions are logged for the post-mortem but no AI is wired. Use the Copy as prompt button below to ask in your own LLM.

you

Is batch B=16 the only decode batch, or can I assume larger B later?

interviewer

(journal mode) Your clarification is logged for the post-mortem. Use the Copy-as-prompt button below to ask in your own LLM.

Ask a clarifying question – e.g. 'what's the latency budget?' or 'how many concurrent users?'

%⌨ to send

Copy as prompt

Ask

Journal mode – nothing leaves your browser. Use Copy-as-prompt and paste into the LLM of your choice if you want answers.